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Development of a recurrence-related gene signature and functional role of MLLT6 in ovarian cancer progression and Paclitaxel resistance

Qiufang Bao^{1,2,3}, Shaoyu Wang^{1,2,3} and Linliang Hong^{4,5*}

Abstract

Objective Ovarian cancer is one of the most lethal gynecological malignancies, with high recurrence rates and poor prognosis. Identifying molecular mechanisms underlying recurrence is critical for improving patient outcomes. This study aimed to develop a recurrence-related gene signature (RRGS) for prognosis and explore the role of Myeloid/Lymphoid or Mixed-Lineage Leukemia; Translocated To 6 (MLLT6), a key gene within the RRGS, in ovarian cancer progression and drug resistance.

Methods TCGA and GEO cohorts were analyzed to identify differentially expressed genes (DEGs) associated with recurrence. LASSO regression was applied to construct an eight-gene RRGS, which was validated in independent cohorts. Functional assays, including proliferation, migration, invasion, and drug resistance tests, were conducted in OVCA3, TOV-21G and ES-2 cell lines to evaluate the role of MLLT6.

Results The RRGS effectively stratified patients into high- and low-risk groups, with high RRGS scores correlating with poorer overall survival and progression-free survival. Among the RRGS genes, MLLT6 was identified as the most significant prognostic marker. Functional assays showed that MLLT6 knockdown inhibited ovarian cancer cell proliferation, migration, invasion, and significantly enhanced Paclitaxel sensitivity by reducing IC50 values.

Conclusion The RRGS is a robust prognostic tool for ovarian cancer, and MLLT6 plays a critical role in tumor progression and drug resistance. Targeting MLLT6 may provide a novel therapeutic strategy to overcome Paclitaxel resistance and improve patient outcomes.

Keywords Ovarian cancer, Recurrence, Immunotherapy, Prognosis, MLLT6

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Introduction

Ovarian cancer remains one of the most lethal gynecological malignancies worldwide [1–3]. Despite advances in surgical techniques and chemotherapeutic regimens, the overall survival rate for ovarian cancer patients has not significantly improved over the past few decades. A major challenge in the management of ovarian cancer is the high rate of recurrence after initial treatment, which is associated with poor prognosis and limited therapeutic options [4]. Understanding the molecular mechanisms underlying ovarian cancer recurrence is essential for developing effective strategies to predict recurrence risk and tailor personalized treatments [5]. Recurrence in ovarian cancer is a complex process influenced by a multitude of genetic and epigenetic alterations [6]. Previous studies have identified various genes and signaling pathways involved in tumor progression, chemoresistance, and metastasis. For instance, dysregulation of genes related to cell cycle control, apoptosis, and DNA repair mechanisms has been implicated in ovarian cancer pathogenesis [7–9]. Additionally, the tumor microenvironment, including immune cell infiltration and cytokine networks, plays a critical role in tumor evolution and response to therapy [10, 11]. However, the specific molecular drivers that distinguish recurrent from non-recurrent ovarian tumors remain inadequately characterized.

Recent advances in high-throughput sequencing technologies have enabled comprehensive profiling of the genomic and transcriptomic landscapes of ovarian cancer. These approaches have revealed significant heterogeneity in gene expression and mutation patterns between recurrent and non-recurrent tumors. Differentially expressed genes (DEGs) and recurrently mutated genes have been implicated in tumor progression, therapy resistance, and immune evasion. However, translating these findings into clinically actionable tools remains a challenge. A systematic approach to integrating gene expression data with clinical outcomes is necessary to identify key molecular drivers of recurrence and develop predictive models for patient stratification.

Therefore, this study aims to elucidate the molecular mechanisms underlying ovarian cancer recurrence by analyzing differential gene expression profiles between followed-up recurrent tissues and followed-up non-recurrent tissues. We seek to identify key DEGs and assess their mutational status to uncover potential drivers of recurrence. Utilizing LASSO regression analysis, we aim to develop an eight-gene RRGs capable of effectively stratifying patients based on recurrence risk and overall survival. Furthermore, we investigate the association between RRGs and the immune microenvironment. Specifically, we focus on the role of MLLT6 within the RRGs, evaluating its prognostic significance across multiple

cohorts. Functional assays are conducted to determine the impact of MLLT6 on ovarian cancer cell proliferation, migration, and invasion. Through this comprehensive analysis, we aspire to provide significant insights into the molecular basis of ovarian cancer recurrence and to propose novel strategies for risk assessment and personalized treatment.

Methods

Data collection

We obtained clinical information and transcriptomic data for 374 ovarian cancer patients from The Cancer Genome Atlas (TCGA) cohort. All tissue samples in the TCGA cohort were collected at the time of initial diagnosis. Based on follow-up data, we categorized 198 patients as “followed-up recurrent cases” (patients who experienced recurrence during follow-up) and 153 patients as “followed-up non-recurrent cases” (patients who did not experience recurrence during follow-up). Additional datasets were sourced from the Gene Expression Omnibus (GEO) database, specifically the GSE53963, GSE8842 and GSE32062 cohorts. The GSE53963 cohort consists of 174 ovarian cancer patients with prognostic information, while the GSE32062 cohort includes 260 ovarian cancer patients with available prognostic data. The GSE8842 cohort includes 73 ovarian cancer patients with prognostic follow-up information.

Differential gene expression analysis

Gene expression levels in both recurrent and non-recurrent ovarian cancer cases were evaluated using the “limma” package in R. Genes exhibiting significant differential expression were identified based on the criteria of $|\log_2 \text{fold change}| \geq 0.585$ and an adjusted p -value below 0.05. These differentially expressed genes (DEGs) were subsequently selected for further downstream analyses.

Functional enrichment analysis

To investigate the biological functions and pathways associated with the identified DEGs, KEGG and GO-BP enrichment analyses were conducted. Pathways and biological processes with an adjusted p -value below 0.05 were considered significantly enriched, highlighting their potential involvement in ovarian cancer recurrence.

Development of recurrence-related gene signature (RRGS)

To create the RRGs, we selected genes with significant prognostic value and applied the LASSO Cox regression model to pinpoint the most reliable prognostic genes. LASSO utilizes L1 regularization, which minimizes the coefficients of less impactful features to zero during the model fitting process. Genes with coefficients reduced to zero were deemed unnecessary and removed from the final model. Using the expression levels of the remaining

genes, we constructed the RRGs formula. The RRGs is calculated as: $RRGs = \sum(\beta_i \times E_i)$, where β_i is the coefficient of the i -th gene, and E_i is the expression level of the i -th gene. The summation includes all selected genes. To determine the effectiveness of RRGs as a prognostic marker for ovarian cancer patients, we performed survival analysis.

Immune microenvironment analysis

To evaluate immune cell infiltration in ovarian cancer samples, we utilized several computational tools, including ESTIMATE, CIBERSORT, and XCELL algorithms. After performing these analyses, we examined the relationships between the RRGs and various immune cell types to understand how the gene signature correlates with the tumor immune landscape.

Cell culture

ES-2, TOV-21G and OVCAR3 cell lines were obtained from the American Type Culture Collection (ATCC). Both cell lines were authenticated by short tandem repeat (STR) profiling prior to use in experiments shown in this work. Upon completion of these studies, both lines were re-authenticated by STR profiling (June 2024) to ensure that no contamination had occurred during these experiments. ES-2 cells were cultured in RPMI-1640 medium supplemented with 10% (v/v) fetal bovine serum (FBS) and 100 μ g/mL streptomycin. OVCAR3 cells were maintained in RPMI-1640 containing 20% FBS and 100 μ g/mL streptomycin. All cell cultures were incubated at 37 °C in a humidified atmosphere with 5% CO₂.

RNA extraction and quantification

Total RNA was isolated from OVCAR3, TOV-21G and ES-2 cell lines using TRIzol reagent (Invitrogen, CA, USA). The extracted RNA was subsequently reverse-transcribed into cDNA with an mRNA reverse transcription kit (Takara, Japan). To quantify MLLT6 mRNA expression, RT-qPCR was performed using specific primers synthesized by Sangon Biotech (Shanghai, China). The primer sequences were as follows. MLLT6-Forward: 5'-G GTGCCTTCATTCCCCTTTG-3'; MLLT6-Reverse: 5'-G CAGGGGTGAGGTCTCCAGT-3'. GAPDH-Forward: 5'-GGTGAAGGTCGGAGTCAAC-3'; GAPDH-Reverse: 5'-CAAATGAGCCCCAGCCTTC-3'. Gene expression levels were determined using the 2^{- $\Delta\Delta$ CT} method.

Cell proliferation assay

Cell proliferation was assessed using the Cell Counting Kit-8 (CCK-8) assay. OVCAR3, TOV-21G and ES-2 cell lines were cultured as single-cell suspensions and diluted to a concentration of 1 $\times$ 10⁴ cells/mL. A total of 2 $\times$ 10³ cells were seeded into each well of a 96-well plate and incubated under standard conditions (37 °C, 5% CO₂).

At 24, 48, and 72-hour intervals, 10 μ L of CCK-8 reagent was added to each well. After incubating for an additional hour, the absorbance was measured at 450 nm using a microplate reader.

Cell migration assay

To evaluate cell migration, transfected OVCAR3, TOV-21G and ES-2 cells were plated in six-well plates at a density of 1 $\times$ 10⁶ cells per well and allowed to grow until nearly confluent. Cells were then subjected to serum starvation to reduce proliferation effects. A straight scratch was made in the cell monolayer using a 10 μ L pipette tip to create a wound. Detached cells were washed away with PBS. Wound closure was monitored at 0 and 48 h using an inverted microscope, and the migration area was quantified with ImageJ software.

Cell invasion assay

Cell invasive capability was measured using Transwell chambers with polycarbonate membranes (Costar; Corning Inc.). Transfected cells were resuspended in serum-free medium at a concentration of 2 $\times$ 10⁴ cells/mL and added to the upper chamber. The lower chamber was filled with complete medium containing 10% FBS to serve as a chemoattractant. After a 24-hour incubation period, cells that had invaded through the membrane were stained and quantified.

Statistical analysis

Data were processed and analyzed using GraphPad Prism 6.0. All in vitro experiments, including cell proliferation, migration, invasion, and drug resistance assays, were independently repeated three times. Comparisons between groups were performed using one-way ANOVA or Student's t-test as appropriate. For survival analysis, the Cox proportional hazards model and Kaplan-Meier method were employed, with the log-rank test used to determine statistical significance. A p -value of less than 0.05 was considered statistically significant throughout the study.

Results

Differential gene expression analysis reveals key genes associated with ovarian cancer recurrence

To identify genes associated with ovarian cancer recurrence, we analyzed the differential gene expression profiles of 198 ovarian cancer patients with recurrence and 153 without recurrence using the TCGA cohort. The heatmap visualization revealed a clear distinction between the two groups, with 143 genes upregulated and 216 genes downregulated in followed-up recurrent tissues compared to followed-up non-recurrent tissues. These findings suggest substantial alterations in gene

expression that may drive the mechanisms underlying ovarian cancer recurrence (Fig. 1A&B).

To gain further insights into the biological significance of the identified upregulated genes, we performed GO and KEGG enrichment analyses. The GO analysis for the 143 upregulated genes highlighted several enriched biological processes. Notably, these processes included the positive regulation of mRNA catabolic processes, skeletal muscle cell differentiation, and cellular amide metabolic processes. The analysis also identified enrichment in biological processes related to nuclear transcribed mRNA poly(A) tail shortening and organonitrogen compound biosynthetic processes. Furthermore, upregulated genes were implicated in amide and peptide biosynthetic processes, as well as the negative regulation of cellular amide metabolic processes (Fig. 1C). These findings suggest that the upregulated genes may foster conditions favorable for tumor growth and recurrence through various metabolic and regulatory pathways. The KEGG pathway analysis associated with these upregulated genes indicated notable enrichments in several pathways, such as sulfur metabolism, oxidative phosphorylation, and thermogenesis. Other relevant pathways included shigellosis, folate biosynthesis, Huntington disease, and butanoate metabolism, alongside the citrate cycle and RNA degradation. The identification of these pathways underscores the complex biological processes that may be dysregulated in ovarian cancer recurrence and points to potential metabolic vulnerabilities (Fig. 1C).

In contrast, the analysis of the 216 downregulated genes provided further insights into the mechanisms that might contribute to ovarian cancer recurrence. The GO-BP analysis revealed that these genes were primarily enriched in processes such as positive regulation of histone deacetylation, covalent chromatin modification, and RNA splicing. Additionally, they were linked to the positive regulation of protein deacetylation and mRNA processing, as well as peptidyl lysine modification and positive regulation of chromatin organization. The response to interleukin 9 was also noted, indicating potential interactions with the immune environment (Fig. 1D). The KEGG pathway analysis for the downregulated genes highlighted their involvement in several critical cancer-related pathways. Significant enrichments were found in lysine degradation, chronic myeloid leukemia, the spliceosome, and both non-small cell and acute myeloid leukemias. Other pathways included renal cell carcinoma, pathways in cancer, EGFR tyrosine kinase inhibitor resistance pathways, and 2-oxocarboxylic acid metabolism.

To better understand genetic alterations associated with ovarian cancer recurrence, we evaluated the mutation landscape of the 359 differentially expressed genes. Among these, the top 20 genes with the highest mutation

rates in ovarian cancer were identified, with KMT2C, DYNC1H1, and ZSWIM8 standing out with mutation frequencies exceeding 20%. Mutations were predominantly missense mutations, with additional alterations such as nonsense mutations and splice site changes observed. These findings suggest that mutations in key DEGs, particularly those with high mutation rates, may contribute to ovarian cancer recurrence (Fig. 1E).

Development and validation of the RRGs

We further analyzed the impact of differentially expressed genes on the prognosis of ovarian cancer patients. Our results indicated that the expression levels of 23 genes were significantly correlated with patient outcomes (Fig. 2A). To refine our analysis, we applied LASSO regression to these genes (Fig. 2B&C), resulting in the development of a Recurrence Related Gene Signature (RRGS). The RRGs model includes 8 genes expressed as follows: $RRGS = (-0.003 \times \text{WARS1 expression}) + (0.423 \times \text{MLLT6 expression}) + (0.021 \times \text{POU4F1 expression}) + (-0.049 \times \text{TRIM38 expression}) + (-0.011 \times \text{BTN3A2 expression}) + (-0.031 \times \text{MARK3 expression}) + (0.021 \times \text{DMWD expression}) + (-0.015 \times \text{YY1 expression}) + (-0.023 \times \text{MZB1 expression})$ (Supplementary Table 1). Kaplan-Meier survival analysis (Fig. 2D) showed that patients with high RRGs values, as predicted by their initial tumor profiles, exhibited significantly poorer overall survival compared to those with low RRGs values. Time-dependent Receiver Operating Characteristic (ROC) curve analysis indicated that the risk score had good predictive accuracy for 1-year, 3-year, and 5-year prognosis, with Area Under the Curve (AUC) values of 0.75, 0.67, and 0.72, respectively (Supplementary Fig. 1A). A heatmap analysis further visualized the expression patterns of the 8 signature genes in RRGs-High and RRGs-Low subtype (Fig. 2E).

To validate the prognostic value of the RRGs, we analyzed two independent GEO cohorts. In the GSE3963 cohort, patients with high RRGs scores exhibited significantly decreased overall survival compared to those with low RRGs scores (Fig. 2F). Time-dependent ROC curve analysis revealed AUC values of 0.71, 0.67, and 0.72 for predicting 1-year, 3-year, and 5-year prognosis, respectively (Supplementary Fig. 1B). Additionally, the GSE32062 cohort corroborated this finding, showing a significant difference in survival outcomes (Fig. 2G). Time-dependent ROC curve analysis revealed AUC values of 0.64, 0.67, and 0.78 for predicting 1-year, 3-year, and 5-year prognosis, respectively (Supplementary Fig. 1C).

We further examined the expression profiles of the eight genes included in the RRGs across ovarian cancer tissues and adjacent normal tissues using the GSE14407

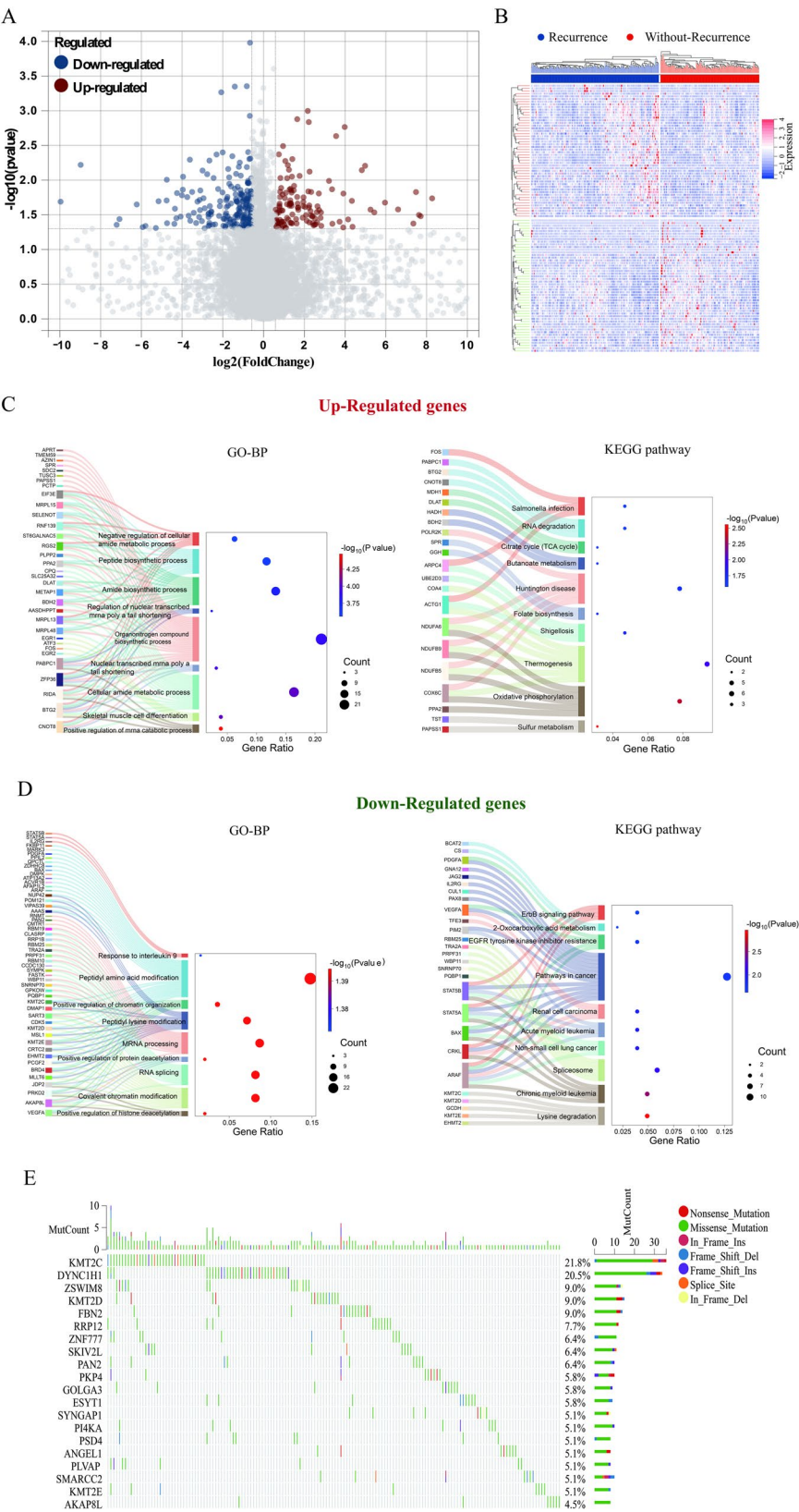


Fig. 1 Differential gene expression and mutation landscape in ovarian cancer recurrence. **(A)** Volcano plot showing the differential gene expression between followed-up recurrent tissues and followed-up non-recurrent tissues in the TCGA cohort. **(B)** Heatmap of gene expression profiles. **(C)** GO and KEGG enrichment analyses of the 143 upregulated genes. **(D)** GO and KEGG analyses of the 216 downregulated genes. **(E)** Mutation landscape of the 359 differentially expressed genes, identifying the top 20 genes with the highest mutation rates

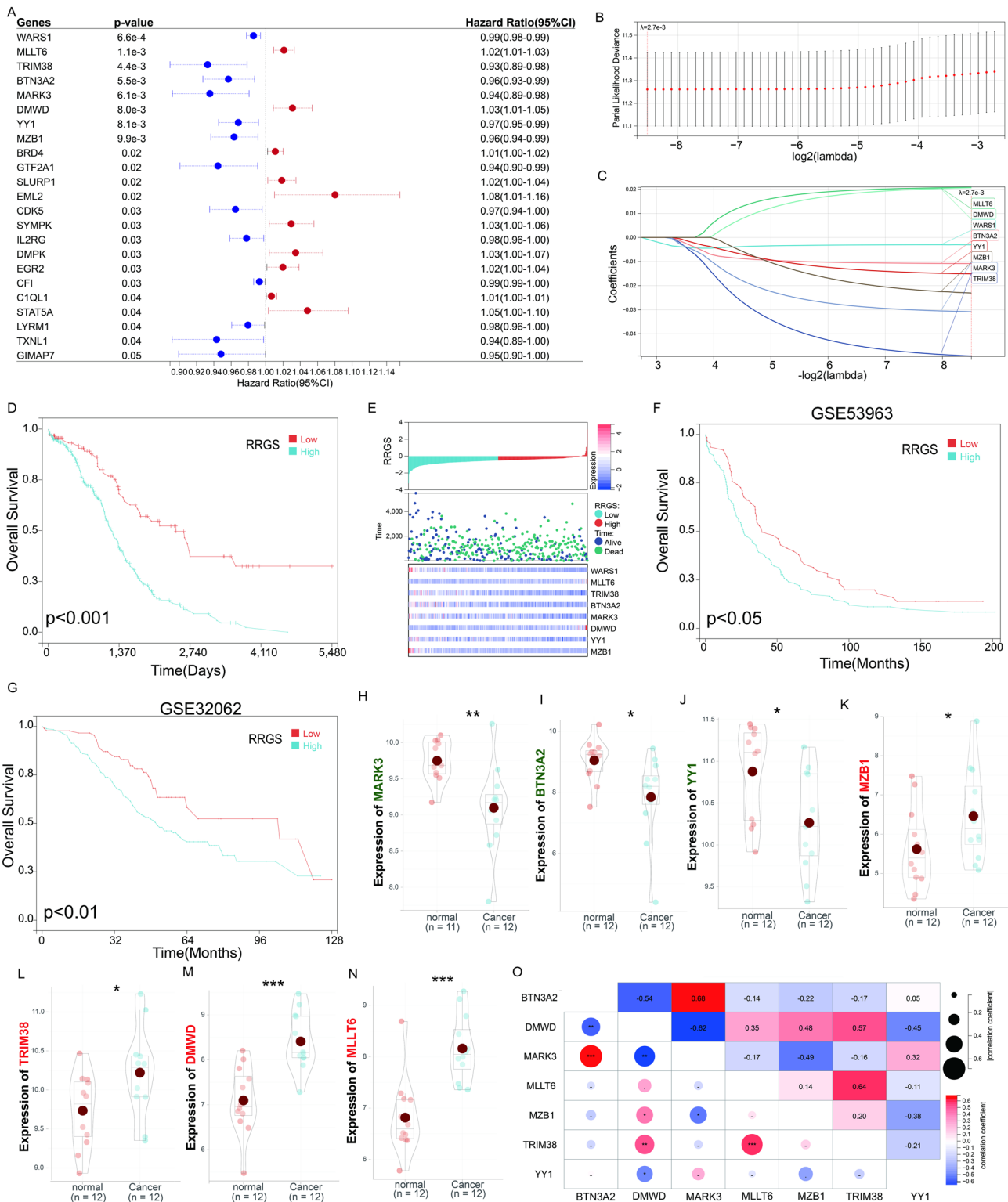


Fig. 2 Development and validation of the RRGs. **(A)** Forest plot illustrating the prognostic significance of 23 differentially expressed genes in ovarian cancer patients. **(B)** LASSO regression profile of the 23 genes with the minimal lambda value marked. **(C)** Coefficient profiles for the selected genes in the LASSO regression model. **(D)** Kaplan-Meier survival curve in the TCGA cohort. **(E)** Heatmap of the expression patterns of the eight RRGs signature genes in RRGs-High and RRGs-Low subtypes. **(F-G)** Kaplan-Meier survival curves validating the prognostic value of the RRGs in the GSE53963 and GSE32062 co-horts. **(H-N)** Violin plots showing the differential expression of the RRGs genes in ovarian cancer tissues versus adjacent normal tissues from the GSE14407 cohort. **(O)** Correlation matrix of the RRGs genes, depicting significant positive and negative correlations. *, $p < 0.05$; **, $p < 0.01$; ***, $p < 0.001$

cohort. The results showed significant differential expression of the genes, with *MARK3*, *BTN3A2*, and *YY1* being notably downregulated in cancerous tissues compared to normal, while *MZB1*, *TRIM38*, *BTN3A2* and *MLLT6* exhibited higher expression levels in cancerous tissues (Figs. 2H-N).

Finally, we conducted a correlation analysis, which revealed significant relationships among the 8 RRGs genes. Notable interactions included positive correlations between *BTN3A2* and *MAPK3*, as well as negative correlations between *DMWD* and *MAPK3* (Fig. 2O). These relationships indicate potential cooperative or antagonistic roles among these genes in ovarian cancer recurrence, providing valuable insights into their functional interplay.

Clinical relevance and mutation landscape of RRGs in ovarian cancer

We further analyzed the clinical relevance of RRGs in ovarian cancer. The results showed no significant differences between RRGs values and patient age or tumor grade (Figs. 3A&B). However, there was a significant association between RRGs values and tumor stage, where higher RRGs values were observed in patients with advanced stages (III-IV) compared to those in early stages (I-II) (Fig. 3C).

GSEA revealed distinct pathway enrichments between RRGs-High and RRGs-Low subgroups. Several biological processes were significantly enriched in the RRGs-High subgroup, including mucociliary clearance, sensory perception of touch, brush border assembly, and positive regulation of protein catabolic processes in the vacuole. Additional enriched pathways involved epithelial-mesenchymal cell signaling and membrane depolarization during SA node cell action potential (Fig. 3D).

The mutational landscape analysis revealed distinct patterns between RRGs subgroups. In the RRGs-Low subgroup, the top 20 mutated genes included *TP53*, *TTN*, *CSMD3*, *RYR2*, *MUC16*, *PRUNE2*, *APOB* and *FLG2* (Fig. 3E). The RRGs-High subgroup exhibited a different mutation profile, with genes such as *TP53*, *TTN*, *CSMD3*, *FAT3*, *FLG*, *USH2A*, *MUC16*, *AHNAK2*, and *LRP1B* among the top mutated genes (Fig. 3F). Comparative analysis between the two subgroups identified several differentially mutated genes. Notably, genes including *SCN5A*, *EGFR*, *NDST1*, *MMRN1*, and *NOTCH3* showed significantly higher mutation rates in the RRGs-High subgroup compared to the RRGs-Low subgroup, with mutation frequencies ranging from 2.3 to 3.5% (Fig. 3G). These findings suggest that specific mutation patterns may contribute to the prognostic differences observed between RRGs subgroups.

Association between RRGs and immune cell infiltration

We further investigated the relationship between RRGs and immune cell infiltration. Using ESTIMATE analysis, we observed that the RRGs-Low subgroup had significantly higher immune scores compared to the RRGs-High subgroup, suggesting an overall higher immune cell infiltration in patients with lower RRGs (Fig. 4A). Additionally, immune subgroup analysis using the CIBERSORT algorithm demonstrated that the RRGs-Low group exhibited significantly higher levels of CD8+ T cells and M1 macrophages, indicating a distinct immune profile between RRGs subgroups (Figs. 4A-B).

To validate these findings, we performed XCELL analysis, which confirmed that the infiltration of CD8+ T cells and M1 macrophages remained significantly higher in the RRGs-Low subgroup compared to the RRGs-High subgroup (Fig. 4C). This consistent observation across multiple analytical approaches highlights a robust association between the RRGs-Low status and elevated immune cell infiltration, particularly with CD8+ T cells and M1 macrophages, both known to play crucial roles in anti-tumor immunity.

We also analyzed the relationship between RRGs and immune checkpoints to further explore the immunological differences between RRGs subgroups. The expression levels of critical immune checkpoint genes, including *PD-L1*, *LAG3*, *PD-1*, and *CTLA4*, were significantly elevated in the RRGs-Low group compared to the RRGs-High group (Fig. 4D). These findings suggest that patients in the RRGs-Low subgroup may exhibit a more active immune environment, potentially contributing to better immunosurveillance but also pointing to possible immune exhaustion due to increased immune checkpoint expression.

MLLT6 as a prognostic marker for ovarian cancer

We employed random forest analysis to evaluate the importance of the 8 RRGs genes in distinguishing between recurrent and non-recurrent ovarian cancer cases. Among these genes, *MLLT6* emerged as the most significant predictor, exhibiting the highest importance score, followed by *MZB1* and *YY1* (Fig. 5A).

Given *MLLT6*'s prominent role, we conducted comprehensive survival analyses across multiple cohorts to validate its prognostic significance. In the GSE8842 cohort, patients with low *MLLT6* expression demonstrated significantly better overall survival compared to those with high expression (Fig. 5B). This finding was further corroborated in the GSE53963 cohort, where low *MLLT6* expression was again associated with improved overall survival (Fig. 5C). Analysis of the TCGA_OV cohort revealed consistent results, with low *MLLT6* expression correlating with significantly better overall survival (Fig. 5D). Furthermore, disease-specific survival analysis

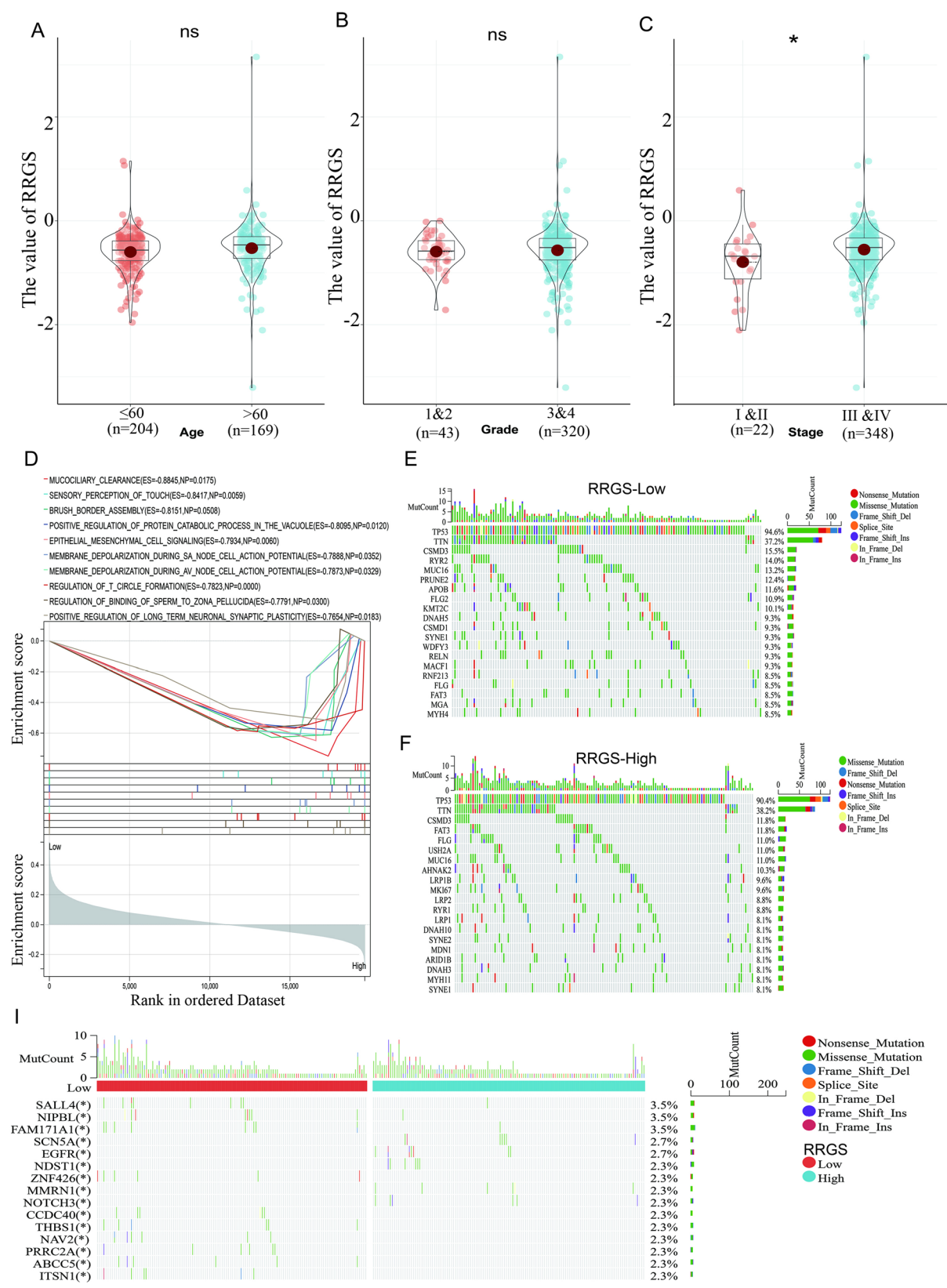


Fig. 3 Clinical relevance and mutation landscape of RRGs in ovarian cancer. **(A)** Association between RRGs values and patient age, **(B)** association between RRGs values and tumor grade, and **(C)** association between RRGs values and tumor stage. **(D)** GSEA of biological pathways enriched in RRGs-High vs. RRGs-Low subgroups. **(E-F)** Waterfall plots of the mutational landscape for RRGs-Low **(E)** and RRGs-High **(F)** subgroups, showing the top 20 frequently mutated genes in each subgroup. **(G)** Comparative analysis of differentially mutated genes between RRGs-High and RRGs-Low subgroups

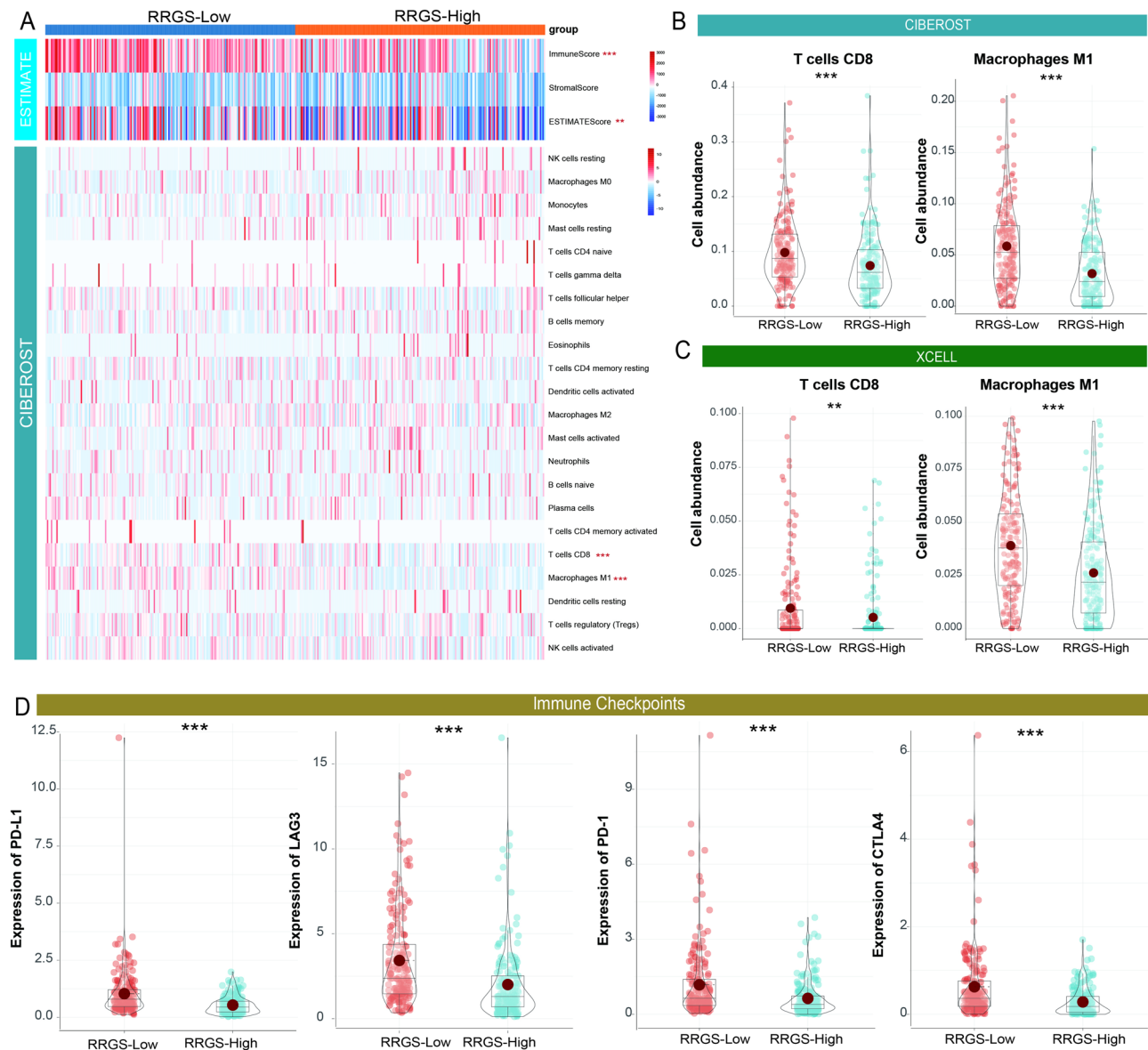


Fig. 4 Association between RRGs and immune cell infiltration in ovarian cancer. **(A)** Heatmap summarizing ESTIMATE and CIBERSORT analyses showing differences in immune scores and immune cell composition between RRGs-Low and RRGs-High subgroups. **(B)** Violin plots from CIBERSORT analysis illustrating significantly elevated levels of CD8+T cells and M1 macrophages in the RRGs-Low subgroup compared to RRGs-High. **(C)** Violin plots from XCELL analysis confirming higher infiltration of CD8+T cells and M1 macrophages in the RRGs-Low subgroup across an independent analytical approach. **(D)** Violin plots showing significantly higher expression levels of immune checkpoint genes, including PD-L1, LAG3, PD-1, and CTLA4, in the RRGs-Low subgroup, compared to RRGs-High. **, $p < 0.01$; ***, $p < 0.001$

in the TCGA_OV cohort showed that patients with low MLLT6 expression had significantly better outcomes (Fig. 5E). Additionally, progression-free survival analysis in the GSE8842 cohort demonstrated that low MLLT6 expression was associated with significantly longer time to disease progression (Fig. 5F).

These consistent findings across multiple independent cohorts and different survival endpoints strongly support MLLT6's role as a crucial prognostic marker in ovarian cancer, suggesting its potential utility as a therapeutic target.

Knockdown of MLLT6 inhibits ovarian cancer cell proliferation, metastasis, and Paclitaxel resistance

To explore the functional role of MLLT6 in ovarian cancer, we performed knockdown experiments in ovarian cancer cell lines. Successful knockdown of MLLT6 expression in OVCAR3 cells was confirmed at the mRNA level (Fig. 6A). Functional assays were then conducted to assess the impact of MLLT6 knockdown on ovarian cancer cell proliferation, migration, and invasion.

In OVCAR3 cells, CCK-8 proliferation assays demonstrated significantly reduced cell viability following

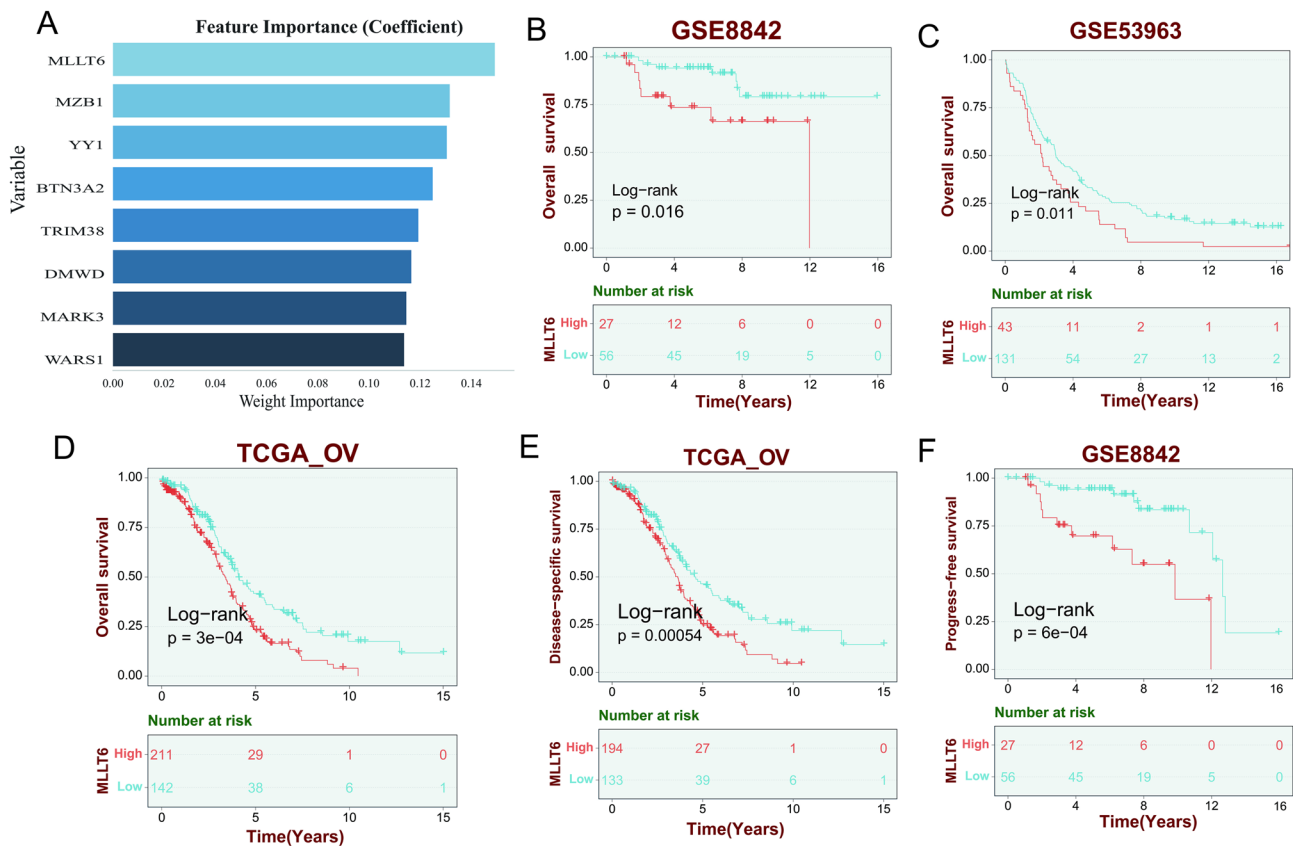


Fig. 5 MLLT6 as a prognostic marker for ovarian cancer. **(A)** Random forest analysis showing the importance scores of the 8 RRGs genes in distinguishing between recurrent and non-recurrent ovarian cancer cases, with MLLT6 identified as the most significant predictor. **(B-C)** Kaplan-Meier overall survival analyses in the GSE8842 **(B)** and GSE53963 **(C)** cohorts. **(D)** Kaplan-Meier overall survival analysis in the TCGA_OV cohort. **(E)** Disease-specific survival analysis in the TCGA_OV cohort. **(F)** Kaplan-Meier progression-free survival analysis in the GSE8842 cohort

MLLT6 knockdown (Fig. 6B). The wound-healing assay revealed a marked reduction in the migration ability of OVCAR3 cells with MLLT6 knockdown compared to control cells (Fig. 6C). Furthermore, transwell invasion assays showed that the invasive capacity of OVCAR3 cells was significantly inhibited after MLLT6 knockdown (Fig. 6D). Consistent results were observed in ES-2 cells, where we further assessed the effects of MLLT6 knockdown (Fig. 6E). Knockdown of MLLT6 in ES-2 cells resulted in reduced cell viability as revealed by the CCK-8 assay (Fig. 6F). The wound-healing assay demonstrated impaired migration ability in MLLT6-knockdown ES-2 cells (Fig. 6G). Moreover, transwell invasion assays confirmed a significant decrease in the invasion ability of ES-2 cells following MLLT6 knockdown (Fig. 6H). Furthermore, we found that MLLT6 knockdown also significantly inhibited the proliferation, migration, and invasion abilities of TOV-21G cells (Fig. 6I-L).

We assessed the functional role of MLLT6 overexpression in ovarian cancer cells. MLLT6 overexpression was confirmed at the mRNA level in OVCAR3 cells (Fig. 6M). In functional assays, overexpression of MLLT6 significantly increased the proliferative capacity of OVCAR3

cells, as demonstrated by the CCK-8 assay (Fig. 6N). The wound-healing assay revealed that MLLT6 overexpression enhanced the migratory ability of OVCAR3 cells, with a marked increase in the migration area compared to the control group (Fig. 6O). Furthermore, transwell invasion assays showed a significant increase in the invasive capacity of OVCAR3 cells following MLLT6 overexpression (Fig. 6P). Similar results were observed in ES-2 cells. MLLT6 overexpression in ES-2 cells was confirmed at the mRNA level (Fig. 6Q). Overexpression of MLLT6 enhanced proliferation, as shown by the CCK-8 assay, with significantly higher cell viability compared to the control group (Fig. 6R). Migratory ability was also significantly increased in MLLT6-overexpressing ES-2 cells, as evidenced by the wound-healing assay (Fig. 6S). In addition, transwell invasion assays revealed that ES-2 cells overexpressing MLLT6 demonstrated increased invasive capacity, indicating a stronger ability to penetrate the extracellular matrix (Fig. 6T). In TOV-21G cells, overexpression of MLLT6 significantly increased the proliferation, migration, and invasion abilities (Fig. 6U-X).

Finally, we examined the role of MLLT6 in modulating drug resistance in ovarian cancer cells. In OVCAR3

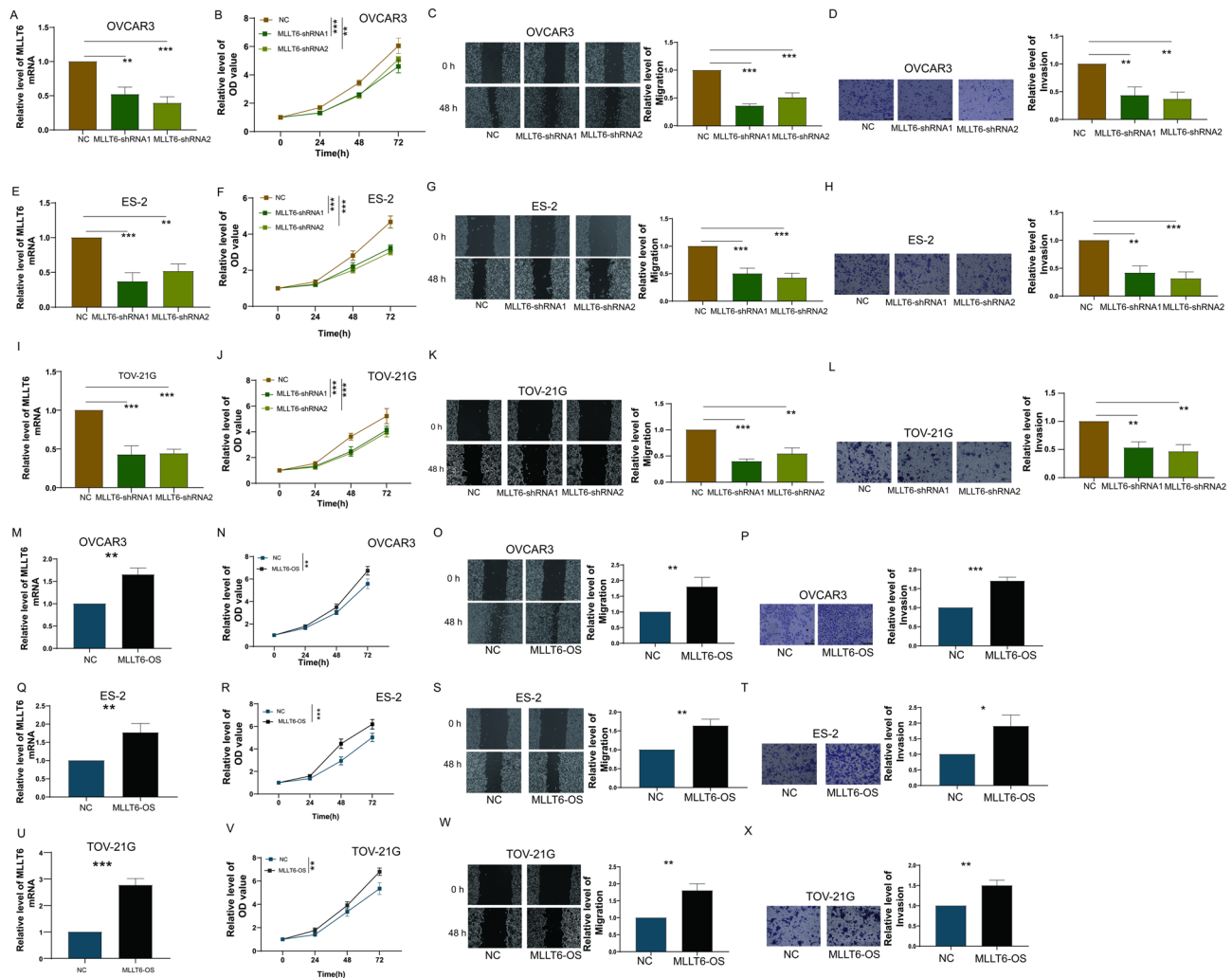


Fig. 6 Functional analysis of MLLT6 in ovarian cancer cells. (A) qRT-PCR validation of MLLT6 knockdown efficiency using two different shRNAs in OVCAR3 cell. (B) Cell proliferation assessed by CCK-8 assay. (C) Wound healing assay showing reduced cell migration at 48 h post-scratch. (D) Transwell invasion assay results. (E) qRT-PCR confirmation of MLLT6 knockdown efficiency in ES-2 cell. (F) Cell proliferation measured by CCK-8 assay. (G) Wound healing assay demonstrating decreased cell migration. (H) Transwell invasion assay results. (I) qRT-PCR confirmation of MLLT6 knockdown efficiency in TOV-21G cell. (J) Cell proliferation measured by CCK-8 assay. (K) Wound healing assay demonstrating decreased cell migration. (L) Transwell invasion assay results. (M) qRT-PCR validation of MLLT6 overexpression efficiency in OVCAR3 cell. (N) Cell proliferation assessed by CCK-8 assay. (O) Wound healing assay showing increased cell migration at 48 h post-scratch. (P) Transwell invasion assay results. (Q) qRT-PCR confirmation of MLLT6 overexpression efficiency in ES-2 cell. (R) Cell proliferation measured by CCK-8 assay. (S) Wound healing assay demonstrating increased cell migration. (T) Transwell invasion assay results. (U) qRT-PCR confirmation of MLLT6 overexpression efficiency in TOV-21G cell. (V) Cell proliferation measured by CCK-8 assay. (W) Wound healing assay demonstrating increased cell migration. (X) Transwell invasion assay results. *, $p < 0.05$; **, $p < 0.01$; ***, $p < 0.001$

cells, the MLLT6 knockdown groups (MLLT6-shRNA1 and MLLT6-shRNA2) exhibited significantly lower IC₅₀ values compared with the NC group (Fig. 7A). Likewise, in ES-2 and TOV-21G cells, MLLT6 knockdown resulted in a substantial reduction in the IC₅₀ values of Paclitaxel (Fig. 7B&C). These findings suggest that silencing MLLT6 markedly enhances Paclitaxel sensitivity in both OVCAR3, ES-2, and TOV-21G cells.

Transcriptome analysis of MLLT6 overexpression in ovarian cancer cells

To investigate the global transcriptional changes induced by MLLT6 overexpression, we performed RNA-seq analysis in OVCAR3 cells overexpressing MLLT6 and control cells. The transcriptome analysis identified a total of 3012 significantly dysregulated genes (1634 upregulated and 1378 downregulated) (Fig. 8A&B, Supplementary Table 2). KEGG pathway analysis of the upregulated genes revealed significant enrichment in Metabolic pathways, MAPK signaling pathway and TNF signaling pathway (Fig. 8C). In addition, GO-BP enrichment analysis

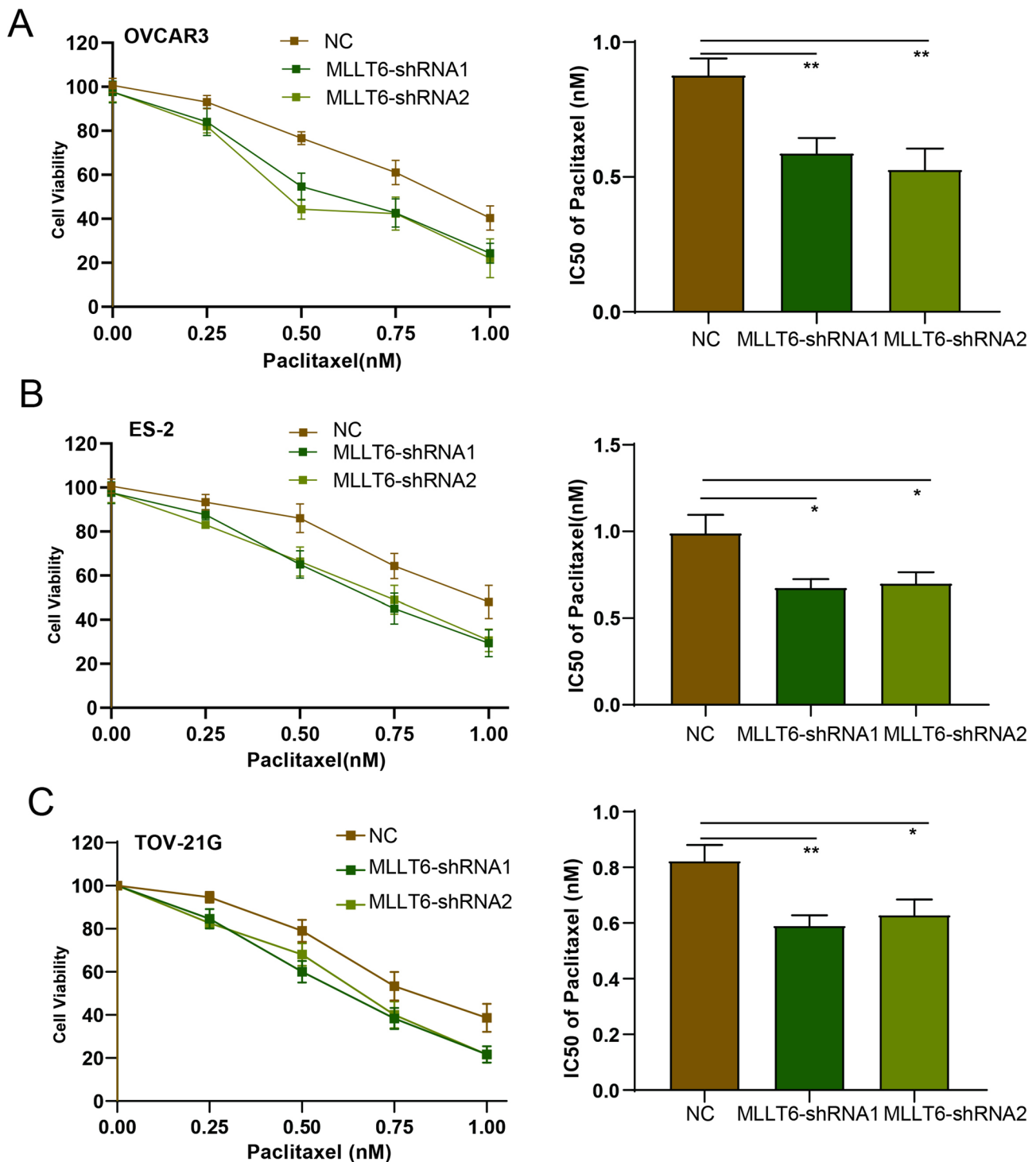


Fig. 7 Knockdown of MLLT6 inhibits ovarian cancer cell Paclitaxel resistance. The IC50 assay shows changes in the drug IC50 of Paclitaxel after MLLT6 knockdown in (A) OVCAR3, (B) ES-2 and (C) TOV-21G cells. *, $p < 0.05$; **, $p < 0.01$

highlighted several biological processes for upregulated genes, including Response to endoplasmic reticulum stress, Response to type I interferon and Small molecule metabolic process (Fig. 8D). For the downregulated genes, KEGG pathway analysis of downregulated genes

indicated significant enrichment in Spliceosome, RNA transport and Cell cycle (Fig. 8E). GO-BP analysis demonstrated enrichment in Ribonucleoprotein complex biogenesis, Ribosome biogenesis and RNA processing (Fig. 8F).

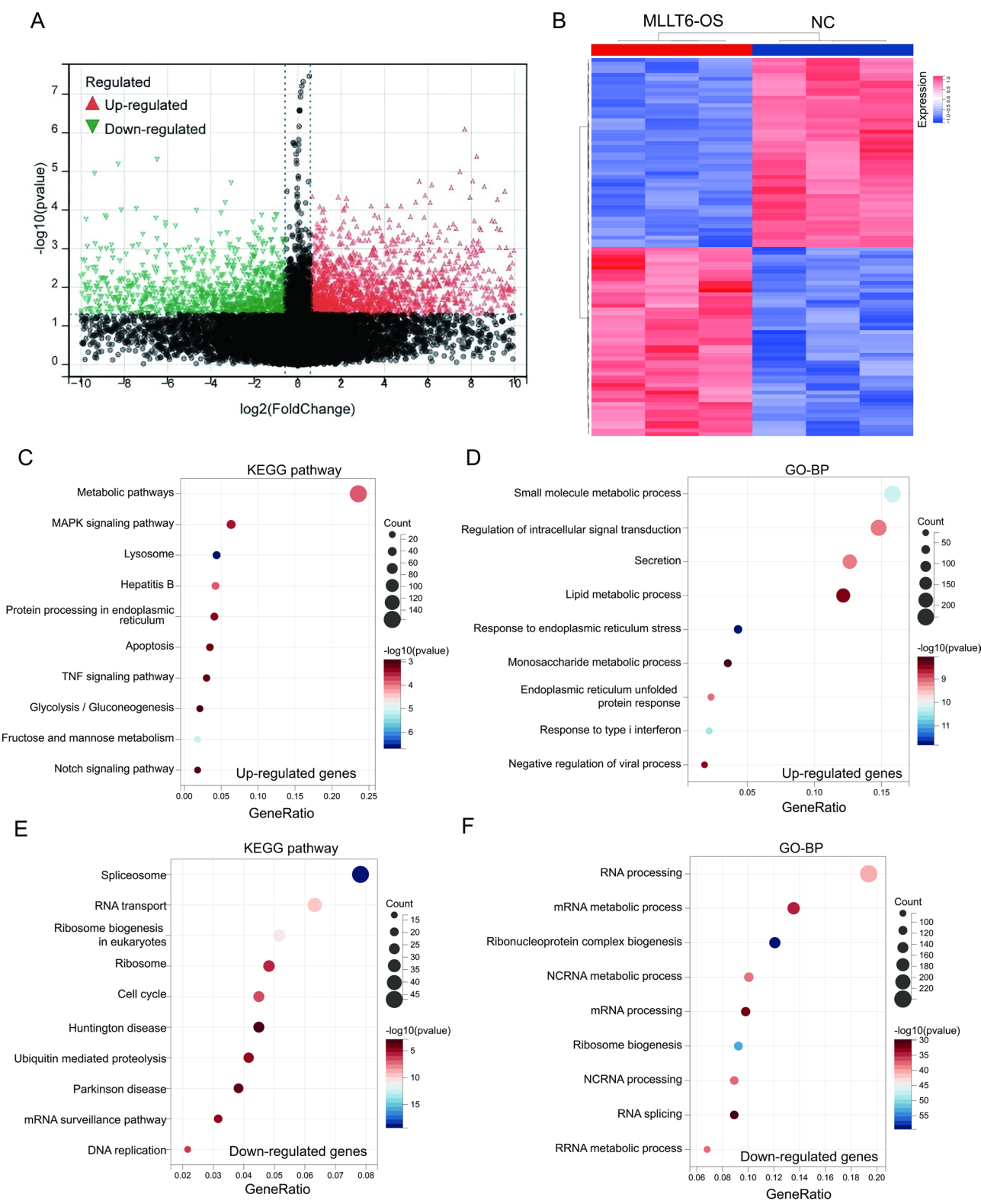


Fig. 8 Transcriptomic analysis of MLLT6-overexpressing OVCAR3 cells. **(A-B)** Transcriptome profiling of OVCAR3 cells overexpressing MLLT6 versus control cells. **(A)** Volcano plot depicting DEGs. **(B)** displays a regulated, upregulated and downregulated distribution of genes upon MLLT6 overexpression. **(C-D)** KEGG **(C)** and GO-BP **(D)** enrichment analyses of genes upregulated in MLLT6-overexpressing cells. **(E-F)** KEGG **(E)** and GO-BP **(F)** enrichment analyses of genes downregulated upon MLLT6 overexpression

Discussion

In this study, we performed a comprehensive analysis to uncover the molecular mechanisms underlying ovarian cancer recurrence and to develop a novel RRGs with significant prognostic and therapeutic implications. Our findings shed new light on the genetic alterations, biological processes, and pathways associated with ovarian cancer recurrence, providing valuable insights for future research and clinical applications. Our RRGs distinguishes itself from previously established gene panels in ovarian cancer through several key aspects. While numerous gene signatures have been developed for ovarian cancer prognosis, our RRGs specifically focuses on recurrence-related mechanisms derived from comparing followed-up recurrent versus non-recurrent tissues. Unlike conventional prognostic panels that primarily predict overall survival, our 8-gene RRGs was specifically designed to stratify patients based on recurrence risk.

Our differential gene expression analysis between followed-up recurrent tissues and followed-up non-recurrent tissues revealed substantial alterations in gene expression profiles. Specifically, we identified 143 upregulated genes and 216 downregulated genes in followed-up recurrent tissues compared to followed-up non-recurrent tissues. These results suggest that ovarian cancer recurrence is associated with a distinct transcriptomic landscape, reflecting changes in biological processes that may drive tumor progression and therapy resistance.

The upregulated genes were enriched in biological processes related to mRNA catabolic processes, amide biosynthesis, and oxidative phosphorylation. The positive regulation of mRNA catabolic processes implies an increased turnover of mRNA molecules, which may facilitate rapid adaptation to changing cellular conditions and promote tumor growth [12]. Additionally, the enrichment in amide and peptide biosynthetic processes indicates heightened protein synthesis activity, supporting the metabolic demands of proliferating tumor cells [13]. The involvement of oxidative phosphorylation suggests that recurrent ovarian cancer cells may rely more heavily on mitochondrial respiration, potentially contributing to increased energy production and resistance to apoptosis. KEGG pathway analysis further highlighted significant enrichments in sulfur metabolism, oxidative phosphorylation, thermogenesis, and the citrate cycle. The upregulation of sulfur metabolism and the citrate cycle pathways may reflect metabolic reprogramming in recurrent tumors, enabling them to cope with oxidative stress and support anabolic growth processes [14]. The identification of pathways such as folate biosynthesis and butanoate metabolism suggests alterations in nucleotide synthesis and short-chain fatty acid metabolism, which could provide precursors for DNA replication and membrane biosynthesis, respectively.

Our evaluation of the mutation landscape among the differentially expressed genes identified several genes with high mutation frequencies, notably KMT2C, DYNC1H1, and ZSWIM8. The predominance of missense mutations suggests that these genetic alterations may lead to functional changes in the encoded proteins, potentially contributing to cancer development and recurrence. KMT2C, a histone methyltransferase, plays a role in epigenetic regulation, and its mutation may disrupt normal chromatin modification processes [15]. DYNC1H1, a cytoplasmic dynein heavy chain, is involved in cellular transport mechanisms, and mutations could affect intracellular trafficking vital for cell division and survival [16]. ZSWIM8 has been less studied, but its high mutation rate warrants further investigation into its potential role in ovarian cancer.

Building on these findings, we developed the RRGs by applying LASSO regression to genes significantly correlated with patient outcomes. The RRGs comprises eight genes: WARS1, MLLT6, POU4F1, TRIM38, BTN3A2, MARK3, DMWD, and YY1. The model effectively stratified patients into high-risk and low-risk groups, with high RRGs values associated with significantly poorer overall survival. The validation of RRGs in independent GEO cohorts (GSE3963 and GSE32062) reinforces its potential as a robust prognostic tool. WARS1 plays a significant role in cancer progression by influencing various cellular processes. In colorectal cancer, higher WARS1 expression correlates with tumor growth, migration, and poor prognosis, making it a potential therapeutic target [17]. Moreover, it facilitates tryptophan-to-phenylalanine substitutions that can enhance antigen presentation, impacting the tumor-immune microenvironment [18]. Additionally, WARS1 has been linked to chemoresistance in breast cancer, where its overexpression predicts a favorable response to chemotherapy [19]. MLLT6 is implicated in various cancers, acting as an oncogene. In acute myeloid leukemia, MLLT6 frequently forms fusion proteins with MLL, leading to aberrant transcriptional regulation and promoting leukemogenesis [20]. Additionally, in breast cancer, MLLT6 regulates PD-L1 expression, enhancing immune evasion and resistance to therapy [21]. Its overexpression is associated with aggressive tumor characteristics, and it plays a role in promoting metastasis through various signaling pathways [22]. Furthermore, MLLT6 has been linked to colorectal cancer progression via its involvement in the Wnt signaling pathway [23]. POU4F1 plays a crucial role in various cancers, acting both as an oncogene and a tumor suppressor. In prostate cancer, elevated POU4F1 expression serves as a prognostic biomarker associated with poor survival outcomes [24]. It has been shown to support lung cancer cell survival and chemoresistance by regulating key pathways [25]. Additionally, POU4F1 contributes to the

resistance of breast cancer cells to trastuzumab by modulating ERK1/2 signaling [26]. Conversely, it functions as a tumor suppressor in thyroid cancer by targeting c-MET/STAT3 signaling [27]. TRIM38 functions as a dual regulator in cancer, exhibiting both oncogenic and tumor-suppressive properties. In colorectal cancer, it inhibits tumor progression by enhancing CCT6A ubiquitination, thereby suppressing the MYC pathway [28]. Additionally, in non-small cell lung cancer, TRIM38 suppresses migration and invasion through the AMPK/NF- κ B/NLRP3 pathway [29]. Conversely, TRIM38 triggers the degradation of GLUT1, restricting tumor growth in bladder cancer [30]. BTN3A2 has critical implications in cancer biology. Its expression is associated with favorable prognosis in lung adenocarcinoma due to enhanced immune infiltration [31]. In stomach cancer, BTN3A2 overexpression supports cancer cell proliferation, migration, and invasion, suggesting its oncogenic role [32]. MARK3 is increasingly recognized as an oncogenic driver across various cancers. In castration-resistant prostate cancer, MARK3 overexpression enhances tumor growth and survival, making it a potential therapeutic target [33]. Additionally, it is involved in endometrial cancer, where high levels correlate with poor prognosis [34]. MARK3 also modulates the Hippo signaling pathway, promoting YAP/TAZ dependent cell proliferation and migration in several cancer types [35]. YY1 is a multifunctional transcription factor implicated in various cancers, acting as both an oncogene and a tumor suppressor. It promotes tumor growth and metastasis in breast cancer by activating the PLAUR gene [36] and regulates the PD-L1 expression in immune evasion mechanisms [37]. YY1 also enhances epithelial-mesenchymal transition in esophageal cancer via the AKT/mTOR pathway [38]. In hepatocellular carcinoma, YY1's interaction with the ERK pathway contributes to cell growth and facilitates cuproptosis, a cancer cell death mechanism [39]. Research on DMWD in cancer is limited, and no direct evidence of its role in cancers has been established.

Clinically, the RRGs was significantly associated with tumor stage but not with patient age or tumor grade. Higher RRGs values were observed in patients with advanced-stage (III-IV) disease, highlighting its potential utility in identifying patients at greater risk of poor outcomes. GSEA revealed that the RRGs-High subgroup exhibited enrichment in biological processes related to EMT, protein catabolism, and sensory perception. The activation of EMT-related pathways may facilitate tumor cell invasion and metastasis, contributing to recurrence.

The mutational landscape between RRGs subgroups showcased distinct patterns. The RRGs-High subgroup had higher mutation rates in genes like SCN5A, EGFR, NDST1, MMRN1, and NOTCH3. Mutations in EGFR and NOTCH3 are particularly noteworthy due to their

known roles in cancer signaling pathways. EGFR mutations can lead to constitutive activation of downstream signaling cascades promoting proliferation [40], while NOTCH3 mutations may affect cell differentiation and survival [41]. These mutations could partially explain the aggressive phenotype observed in patients with high RRGs values.

The analysis of immune cell infiltration revealed that the RRGs-Low subgroup had significantly higher immune scores, indicating a more robust immune presence within the tumor microenvironment. Specifically, increased infiltration of CD8⁺ T cells and M1 macrophages was observed. CD8⁺ T cells are cytotoxic lymphocytes crucial for anti-tumor immunity [42], while M1 macrophages are pro-inflammatory and can enhance anti-tumor responses [43]. The elevated expression of immune checkpoint genes in the RRGs-Low subgroup suggests a pre-existing "hot" tumor microenvironment that may be particularly amenable to immune checkpoint blockade. Given that MLLT6 drives both tumor aggressiveness and paclitaxel resistance, combined inhibition of MLLT6 and blockade of PD-1/PD-L1 or CTLA-4 pathways could synergistically suppress tumor growth while reinvigorating exhausted T cells. Preclinical studies are warranted to evaluate whether MLLT6 targeting enhances the efficacy of checkpoint inhibitors in ovarian cancer.

Among the RRGs genes, MLLT6 emerged as the most significant predictor of ovarian cancer recurrence and prognosis. Consistently across multiple cohorts, high MLLT6 expression was associated with poorer overall survival, disease-specific survival, and progression-free survival. This consistent prognostic significance highlights MLLT6 as a potential key player in ovarian cancer biology.

Functional assays provided further evidence of MLLT6's role in promoting ovarian cancer progression. Knockdown of MLLT6 in OVCAR3, TOV-21G and ES-2 cell lines led to reduced cell proliferation, migration, and invasion, indicating that MLLT6 supports the aggressive behavior of ovarian cancer cells. Conversely, overexpression of MLLT6 enhanced these malignant phenotypes, reinforcing its oncogenic potential. Our functional experiments revealed that MLLT6 knockdown significantly reduced IC50 values for Paclitaxel in three different ovarian cancer cell lines. While MLLT6 has been implicated in various cancer types, its specific role in ovarian cancer Paclitaxel resistance represents a novel finding with distinct mechanistic insights. Previous studies have primarily focused on MLLT6's role in immune resistance through PD-L1 regulation and its function in acute myeloid leukemia as part of MLL fusion proteins. However, our study is the first to demonstrate MLLT6's direct

involvement in Paclitaxel sensitivity in ovarian cancer cells.

To further elucidate the molecular mechanisms underlying MLLT6's oncogenic functions in ovarian cancer, we performed comprehensive RNA-seq analysis on OVCAR3 cells overexpressing MLLT6. The transcriptomic profiling revealed 3012 significantly dysregulated genes, with 1634 upregulated and 1,378 downregulated genes. KEGG pathway analysis of the upregulated genes revealed significant enrichment in several cancer-related pathways, including metabolic pathways, the MAPK signaling pathway, and the TNF signaling pathway.

The enrichment of metabolic pathways suggests that MLLT6 overexpression drives extensive metabolic reprogramming in ovarian cancer cells, enabling them to meet the heightened biosynthetic and bioenergetic demands associated with rapid proliferation and metastasis. MAPK pathway dysregulation is a hallmark of cancer progression and is frequently associated with increased cellular motility and invasion capacity. Furthermore, the upregulation of TNF signaling pathway components suggests that MLLT6 may modulate inflammatory responses and promote a pro-survival cellular environment. TNF signaling can contribute to chemoresistance by activating anti-apoptotic pathways and promoting tumor cell survival under therapeutic stress. Collectively, these findings position MLLT6 as a central regulator that coordinates metabolic adaptation, mitogenic signaling, and inflammatory responses to drive ovarian cancer progression and therapeutic resistance.

Our study has several strengths, including the use of large, publicly available datasets for comprehensive analysis and the validation of findings across multiple independent cohorts. The integration of gene expression, mutation data, and immune profiling provides a holistic view of the factors contributing to ovarian cancer recurrence.

However, there are several limitations to consider. The study is retrospective in nature, relying on existing datasets, which may introduce selection bias. The heterogeneity of the datasets, including differences in patient populations, treatment modalities, and data collection methods, could influence the results. An important limitation of our study is the inability to perform direct comparisons between matched primary and recurrent ovarian cancer tissues due to the limited availability of paired samples with complete gene expression data in the TCGA database. This prevents us from identifying recurrence-specific genetic alterations that distinguish primary tumors from their corresponding recurrent counterparts, which would provide more precise insights into the molecular mechanisms driving ovarian cancer recurrence. Moreover, while functional assays in cell lines support the role of MLLT6 in ovarian cancer progression,

in vivo studies and clinical validation are necessary to confirm its therapeutic potential. Future studies should focus on validating the RRGs and MLLT6 as prognostic and predictive biomarkers in prospective clinical trials with larger, more diverse patient cohorts. Investigating the molecular mechanisms underlying MLLT6's function could uncover additional therapeutic targets and enhance our understanding of ovarian cancer biology. Furthermore, exploring the interaction between RRGs genes and the immune microenvironment may reveal insights into how tumors evade immune surveillance and resist immunotherapies. Importantly, future research should prioritize the collection and analysis of matched primary-recurrent tissue pairs to better understand the temporal evolution of genetic alterations during ovarian cancer progression and recurrence.

Conclusion

Our study provides significant insights into the molecular underpinnings of ovarian cancer recurrence. The development of the RRGs offers a valuable tool for risk stratification and prognosis, potentially guiding clinical decision-making. The identification of MLLT6 as a key prognostic marker and potential therapeutic target holds promise for improving outcomes in ovarian cancer patients. Continued research in this area is critical to translating these findings into clinical applications that can enhance patient care and survival.

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s13048-025-01791-3>.

Supplementary Material 1

Supplementary Material 2

Supplementary Material 3

Acknowledgements

Not applicable.

Author contributions

Qiufang Bao and Shaoyu Wang carried out the bioinformatic analyses. Qiufang Bao and Shaoyu Wang conducted the experiments. Linliang Hong supervised the work and wrote the manuscript. All authors reviewed the manuscript.

Funding

This work was supported by Fujian Provincial Health Technology Project (2023TG012), Fujian Provincial Finance Project (BPB-2024HLL), and Natural Science Foundation of Fujian Province (2022J01218).

Data availability

No datasets were generated or analysed during the current study.

Declarations

Ethics approval and consent to participate

Not applicable.

Competing interests

The authors declare no competing interests.

Clinical trial number

Not applicable.

Received: 30 January 2025 / Accepted: 18 August 2025

Published online: 15 October 2025

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